general term in the state of the art, the question arises whether the general term makes the claimed matter fully or partially accessible to the public. In other words, it has to be established whether the general term used in the citation discloses the subject-matter defined by the special term in the claim. The prior-art disclosure needs to be identified particularly carefully in such cases. General terms of this kind occur particularly frequently in the chemical literature, which is why the relevant case law usually relates to this field (see e.g. **T 2350/16**, concerning an application in the field of Mechanics, in which the board observed that the case law on selections from lists could not be applied because it did not concern (long) lists of the kind commonly encountered in chemistry but, in each instance, merely a selection from among two or three elements at most; see also **T 712/16**). There are two types of case here:

(a) assessing the novelty of chemical substances and groups of substances in respect of general formulae (Markush formulae) under which they fall (see chapter I.C.6.2. "Novelty of chemical compounds and groups of compounds"), and

(b) assessing the novelty of products or processes defined by parameter ranges as against known products or processes characterised by wider or overlapping parameter ranges (see chapter I.C.6.3. "Selection of parameter ranges").

These types differ mainly in technical terms, but the same principles of patent law apply to both. For this reason, the boards of appeal have always been able to adopt the same approach to questions of this nature.

6.2. Novelty of chemical compounds and groups of compounds

According to the boards' case law, a specific combination of elements requiring the selection of elements from two known groups/lists cannot be regarded as disclosed in the art and so fulfils the novelty requirement (cf. **T 12/81**, OJ 1982, 296).

In the landmark decision **T 12/81** (OJ 1982, 296) is stated that the teaching of a cited document is not confined to the detailed information given in the examples of how the invention is carried out, but embraces any information in the claims and description enabling a person skilled in the art to carry out the invention. If a product cannot be defined by a sufficiently accurate **generic formula**, it is permissible to make the definition more precise by additional product parameters such as melting point, hydrophilic properties, NMR coupling constant or the method of preparation (product-by-process claims). From this it necessarily follows that patent documents using such definitions will be prejudicial to the novelty of later applications claiming the same substance defined in a different and perhaps more precise way. Summarising, the board stated that in the case of one of a number of chemical substances described by its structural formula in a prior publication, the particular stereo-specific configuration of the substance – though not explicitly mentioned – was disclosed in a manner which was prejudicial to novelty, if it proved to be the inevitable but undetected result of one of a number of processes adequately described in the prior publication by the indication of the **starting compound** and the **process**.

The applicant argued that the novelty of the claimed product was based on a selection. The board used the opportunity to comment on this argument and develop **criteria for selection inventions** that have frequently been adopted in later decisions: A substance selection can come about if an unmentioned compound or group of compounds having a formula covered by the state of the art is found, in the absence of any information as to the starting substance or substances. The subject-matter in the case in question, however, did not involve a selection of that kind in an area which, although marked out by the state of the art, was nonetheless virgin territory. However, the disclosure by description in a cited document of the starting substance as well as the reaction process is always prejudicial to novelty because those data unalterably establish the end product. If, on the other hand, two classes of starting substances are required to prepare the end products, and examples of individual entities in each class are given in two lists of some length, then a substance resulting from the reaction of a specific pair from the two lists can nevertheless be regarded for patent purposes as a selection and hence as new.

The board held that a combination of starting substances and process variants, however, was quite a different matter from a combination of two starting substances, and thus not comparable. At its simplest, if the starting substances were regarded as fragments of the end product, then every conceivable combination of a given starting substance in the first list with any starting substance in a separate second list of additionally required starting substances involved a true substantive modification of the first starting substance, since in every combination it was supplemented by a different fragment of the second starting substance to become a different end product. Each end product was thus the result of two variable parameters. However, combining a given starting substance from a list of such substances with one of the given methods of preparation did not result in a real substance alteration of the starting substance but only an "identical" alteration (see also **T 3/89**, **T 1841/09**).

6.2.1 Anticipation of certain compounds

a) Definition of a substance by its structural formula or other parameters

Following **T 12/81** (see above) the board in **T 352/93** decided that a claim for an ionic compound (salt) that was defined only by structural parameters, i.e. the structural formulae of the cation and anion of the compound, was not novel over prior art disclosing an aqueous solution that contained a base corresponding to the cation and an acid corresponding to the anion.

In **T 1336/04** the board stated that, according to case law, (see, inter alia, **T 767/95** and **T 90/03**), the preparation of an enzyme sufficiently pure to allow sequencing was novel over a preparation which was not in such a state of purity.

In **T 767/95** concerned the purification of interleukin-1Beta (IL-1Beta), a high molecular weight protein (17.5 kDa). The board found that a purified homogeneous preparation of IL-1Beta was novel over a semi-purified mixture of proteins containing IL-1Beta. A relevant consideration was the provision of IL-1Beta in a degree of purity that allowed the determination of its (partial) amino acid sequence, whereas "no analysis of the amino acid sequence of IL-1 that would provide definitive proof of the homogeneity of IL-1 preparations" was found in the prior art (see also **T 90/03**, **T 29/05**).

b) Selection of starting substances from different lists

According to **T 12/81** (OJ 1982, 296), an end product resulting from the reaction of a specific pair of starting substances may be seen as a novel selection for patent purposes if its preparation requires using entities from two classes of starting substances given in two lists of some length. This criterion has been applied to mixtures of two substances, selected from two lists (**T 401/94**) and confirmed in subsequent decisions (**T 211/93**, **T 175/86**, **T 806/02**, **T 2436/10**).

In **T 401/94** the board again adopted one of the criteria for selection inventions laid down in decision **T 12/81** (OJ 1982, 296), namely, that if two classes of starting substances were required to prepare the end products, and examples of individual entities in each class were given in two lists of some length, the substance resulting from the reaction of a specific pair from the two lists could be regarded for patent purposes as a selection and, hence, as new. The board stated that, although **T 12/81** concerned the synthesis of a chemical product, and the case in question involved the preparation of a **mixture**, the claimed subject-matter was defined on the basis of two chemical entities, each of which had been selected from a list of compounds. Hence the criteria defined in **T 12/81** were applicable here too. By analogy, the board held that, in this case, the claimed composition had to be viewed as a selection, and therefore as novel, as it corresponded to a specific combination of constituents, each of which had been selected from a relatively long list.

In **T 366/96** the board held that if, when selecting two components of a composition from two known lists of possible ingredients, a skilled person had, as soon as one component was taken from the first list, no choice in selecting the second component from the second list in view of compelling technical necessities which made the particular second component mandatory, then this could not be considered to be a "twofold" selection which could render the resulting combination novel.

In **T 754/10** the board came to the conclusion that contrary to the appellant's arguments, the novelty objection was not based on an unallowable selection from different lists, namely the presence or absence of a coating layer combined with the ranges for granule size, core size, total enzyme content and percentual coating content. In fact, there were no lists at all in prior art D15, but rather ranges of values for granule and core sizes and enzyme and coating contents. The need to refer to distinct parts of D15 derived from the fact that the granules as claimed and the granules of D15 were defined by different parameters. A further argument of the appellant was that none of D15's examples fell within the scope of claim 1; in particular they did not work within the area of overlap of the enzyme-content range. Hence the skilled person would not seriously contemplate working within this area of the disclosed range, and this was also in line with the usual practice in the field, which favoured the use of less concentrated enzyme (thus lower percentual enzyme content) in e.g. powder formulations. As regards to these arguments, the board noted that novelty is to be assessed vis-à-vis the whole disclosure of a prior art document, the examples being only a part thereof: it is therefore sufficient that the general part of the description of a prior art document discloses embodiments which are novelty destroying, even if the embodiments disclosed in the examples are not. Claim 1 of the main request lacked novelty over D15.

c) Selection on the basis of a general formula

In **T 181/82** (OJ 1984, 401) the board confirmed that the products of processes which were the inevitable result of a prior description of the starting materials and the process applied thereto formed part of the state of the art. This was true even if one of the two reactants manifested itself as a chemical entity (C_1 alkyl bromide) from a group of generically defined compounds (C_1 - C_4 alkyl bromides). The board took the view that the description of the reaction of a certain starting material with C_1 to C_4 alkyl bromides disclosed only the C_1-substituted product, and was not prepared to recognise the disclosure of a particular butyl substituent on the grounds that four isomeric butyl radicals existed.

In **T 7/86** (OJ 1988, 381) the board also based its reasoning on **T 12/81** (OJ 1982, 296), stating that the principle that a substance resulting from the reaction of a specific pair from two lists could nevertheless be regarded as new, was applicable not only to starting substances in chemical reactions but also to polysubstituted chemical substances where the individual substituents had to be selected from two or more lists of some length, such as in the case in question.

Following on from **T 181/82** (OJ 1984, 401) it was stated in **T 7/86** that if a class of chemical compounds precisely defined only in structural terms (by a chemical reaction), and with only one generically defined substituent, did not represent a prior disclosure of all the theoretical compounds encompassed by an arbitrary choice of a substituent definition, this clearly also had to be the case for a group of chemical substances, the general formula of which had two variable groups. Therefore, a class of chemical compounds defined only by a general structural formula having at least two variable groups did not specifically disclose each of the individual compounds which would result from the combination of all possible variants within such groups.

In **T 258/91** the case concerned a selection from two lists of starting compounds. The compound (formula VI) cited as taking away novelty from the patent in suit differed from the claimed compound (formula I) by the methyl residue on the amino group in the 4-position. In the board's judgment, the information in the cited document was not sufficient to disclose the compound of formula I to the skilled person in the form of a concrete, reproducible technical teaching. The board found that the cited document did not contain any teaching involving the modification of the compound, which was mentioned only by way of example. What was being taught was merely the preparation of a class of compounds and not of a specific, individual compound.

In **T 658/91** the board held that the case law did not suggest that a chemical compound was deemed to be specifically disclosed only if that compound was mentioned by name or even described in an example. On the contrary, it was sufficient if the compound could be unambiguously identified as envisaged in individualised form in the document in question, since the purpose of Art. 54(2) EPC 1973 was to exclude the state of the art from patentability.

6.2.2 Novelty of groups of substances

The case law on the novelty of generically defined compounds and particular examples of these was summarised in decision **T 12/90**. The board had to consider the novelty of a vast family of chemical compounds defined by a general structural formula, where the prior art also disclosed a vast family likewise defined by a general structural formula, the two families having a large number of products in common. The board pointed out that a distinction had to be drawn between two situations:

(a) If the subject-matter of the invention was a particular compound, whereas the prior art disclosed a family of compounds defined by a general structural formula including this particular compound but not describing it explicitly, the invention had to be considered novel (see **T 7/86**, **T 85/87**, **T 133/92**).

(b) If, with the same prior art, the subject-matter of the invention was a second family of compounds partially covering the first, the invention was not new (see **T 124/87**).

As regards case (a) the board said: "That case is not comparable with the present one in which a distinction must be drawn between the novelty of a group of substances defined by a general formula and a second group of substances partially covering the first and defined by another general formula, because the **concept of individualisation** naturally only applies to the structural definition of a single compound, not a collection of compounds".

Case (b) was extensively discussed in **T 124/87** (OJ 1989, 491). This decision dealt with the problem of assessing the novelty of a class of compounds defined by parameters within numerical ranges. The patent in suit claimed a class of compounds defined by parameters within numerical ranges while the prior document disclosed a process by which a class of compounds could be prepared – comprising those claimed in the patent in suit – having the combination of parameters required by the main claim of the latter. In that particular case, the example specifically described in the prior document did not disclose the preparation of any particular compounds within the class defined in the claims of the disputed patent. However, it had been accepted by the patentee that a skilled man would have no difficulty in preparing such compounds within the class defined by the claims of the disputed patent using the process described in the said prior document, in combination with his common general knowledge, so that the disclosure of the prior document had to be regarded as not only limited to the particular compounds whose preparation was described in the examples, but also as comprising the general class of compounds made available to the skilled man in that technical teaching, even though only certain compounds within this class were described as having been prepared. Since the compounds as defined in the claims of the disputed patent formed a major part of this general class, they formed part of the state of the art and therefore lacked novelty.